

PyEnv Installation and Python Environment Setup (Windows)

This report summarizes the installation of PyEnv for Windows and the creation of a Python development environment. PyEnv is a version manager that allows multiple Python versions to be installed and switched easily.

Steps Performed

1. Opened Windows PowerShell as Administrator.
2. Configured execution policy for the current user.
3. Downloaded and ran the PyEnv installation script.
4. Temporarily configured PYENV and PATH environment variables.
5. Verified installation using 'pyenv --version'.
6. Checked available Python versions with 'pyenv versions'.
7. Set a global and local Python version.
8. Created a virtual environment using 'python -m venv .venv'.
9. Activated the virtual environment using '.\venv\Scripts\Activate'.
10. Verified the active Python executable.

Key Commands Used

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
Invoke-WebRequest ... install-pyenv-win.ps1
& ".\install-pyenv-win.ps1"
pyenv --version
pyenv versions
pyenv global <version>
pyenv local <version>
python -m venv .venv
.\venv\Scripts\Activate
where python
python -c "import sys; print(sys.executable)"
```

Result

The environment was successfully configured. PyEnv became available, Python versions could be managed, and a virtual environment was created and activated. This setup is useful for software development because it keeps project dependencies isolated and allows different projects to use different Python versions.

Installation Screenshot

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\shiva> Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
PS C:\Users\shiva> Invoke-WebRequest -Uri "https://raw.githubusercontent.com/pyenv-win/pyenv-win/master/pyenv-win/install-pyenv-win.ps1" -OutFile ".\install-pyenv-win.ps1"
PS C:\Users\shiva> & ".\install-pyenv-win.ps1"
pyenv-win 3.1.1 installed.
No updates available.
PS C:\Users\shiva> $env:PYENV = "$HOME\.pyenv\pyenv-win"
PS C:\Users\shiva> $env:Path = "$env:PYENV\bin;$env:PYENV\shims;$env:Path"
PS C:\Users\shiva> pyenv versions
* 3.10.11 (set by C:\Users\shiva\.pyenv\pyenv-win\version)
  3.12.10
PS C:\Users\shiva> pyenv global 3.10.11
PS C:\Users\shiva> pyenv local 3.10.11
PS C:\Users\shiva> python -m venv .venv
PS C:\Users\shiva> .\.venv\Scripts\Activate
(.venv) PS C:\Users\shiva> where python
(.venv) PS C:\Users\shiva> python -c "import sys; print(sys.executable)"
C:\Users\shiva\.venv\Scripts\python.exe
(.venv) PS C:\Users\shiva>
```